

Rajesh Botla

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LinkedIn

GitHub

Portfolio

Guntur, Andhra Pradesh, India

Professional Summary

Computer Science graduate and **Full-Stack Software Engineer** with proven expertise architecting end-to-end distributed systems and applied AI applications. Proficient in engineering robust backend architectures using **Java (Spring Boot)** and **Python**, seamlessly integrated with interactive frontends via **JavaScript**, **HTML5/CSS3**, and **Streamlit**. Demonstrated success implementing event-driven streaming queues (**Apache Kafka**), secure cloud storage (**AWS S3**), high-performance caching (Redis), and **dual-mode Generative AI pipelines (RAG)**, all grounded in rigid **DSA**, **OOP**, and **clean-code design principles**.

Technical Skills

Languages & Scripting: Python, Java 17, JavaScript (Vanilla/Node.js), SQL, HTML5, CSS3, C

APIs & Integration: RESTful APIs, API Integration, HTTP Methods, API Authentication, Postman

Frameworks & Libraries: Spring Boot, LangChain, Streamlit, Hugging Face Transformers, HLS.js, PyTest, JUnit

Databases & Infrastructure: MySQL, MongoDB (NoSQL), Redis (Caching), AWS S3, Apache Kafka, Vector Databases (FAISS),

Architectural & Design Patterns: Microservices, Event-Driven Architecture, Retrieval-Augmented Generation (RAG), System Design

Tools & DevOps: Docker, Docker Compose, Git, GitHub, Linux/Bash, FFmpeg, OpenCV, Tesseract OCR, pdf2image, PyPDF

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Technical Debugging & Root Cause Analysis, SOLID Principles

Work Experience

AI & ML Engineer Intern – Blackbucks Education Pvt. Ltd.

Dec 2025 – Apr 2026

Guntur, India

- Engineered end-to-end Python ML and data-transformation pipelines using NumPy and Pandas across 5+ datasets, reducing manual data preprocessing and workflow overhead by ~35%.
- Built custom utility scripts for data validation, image feature extraction, and automated preprocessing in Google Colab to accelerate testing cycles by 40%.
- Built a multi-object detection computer vision model with OpenCV, achieving 88% localization accuracy across 10 object classes in real-time video streams.
- Collaborated in an Agile/Scrum engineering group following strict Git version control workflows, participating in routine code audits, cross-functional mapping, and SDLC optimization cycles.

Projects

Netflix-Inspired Distributed Video Streaming Platform | *Java 17, Spring Boot, Kafka, Redis, AWS S3, Docker, REST APIs, Postman*

- Architected a Netflix-inspired streaming platform as four independent Spring Boot microservices (Content, Video, Encoding, Streaming), enabling independently deployable, loosely coupled services with structured REST endpoints.
- Engineered an event-driven processing pipeline with Apache Kafka to trigger asynchronous encoding and streaming workflows, handling data payload transitions across decoupled services.
- Automated video transcoding with FFmpeg to generate adaptive HLS streams in 1080p, 720p, 480p, and 360p for smooth adaptive-bitrate playback.
- Integrated AWS S3 for scalable file storage, MySQL for persistent database management, and Redis to cache metadata, performing root-cause debugging to reduce redundant database reads by over 50%.
- Built a responsive frontend in HTML, CSS, and vanilla JavaScript with HLS.js, consuming REST APIs for real-time video browsing, search, and adaptive streaming.
- Containerized all four microservices, MySQL, Redis, Kafka, and Zookeeper via Docker Compose, enabling reliable deployment and streamlined local integration testing.

SparkAI: LLM-Powered AI Assistant with Document-Based Q&A | *Python, Streamlit, LangChain, Gemini, Groq, FAISS, Docker, OCR*

- Developed a dual-workflow conversational assistant executing clear branches: zero-document General AI Chat and complex Document-Based Question Answering (Q&A).
- Integrated Google Gemini and Groq APIs using LangChain to manage LLM orchestration, conversation flows, and dynamic application query routing.
- Engineered a robust Retrieval-Augmented Generation (RAG) pipeline leveraging Hugging Face Sentence Transformers (*paraphrase-MiniLM-L3-v2*) and a FAISS database for semantic vector similarity search.
- Built an automated document ingestion framework utilizing *RecursiveCharacterTextSplitter* for targeted text chunking/overlap, integrating *PyPDF* alongside a fallback OCR pipeline (*pdf2image* and *pytesseract*) for scanned file extraction.
- Designed a responsive web interface using Streamlit, featuring local session state management, transcript exporting, secure document hashing to eliminate redundant re-embeddings, and complete Docker containerization.

Achievements

- Best Performer (IoT Hackathon):** Awarded Best Performer in a 36-hour IoT Hackathon for building an automated environmental monitoring prototype using microcontrollers, sensors, and threshold alert logic to deliver real-time telemetry.

Education

KITS Akshar Institute of Technology – B.Tech in CSE (CGPA: 7.6/10.0)

Guntur, India – Oct 2022 – June 2026

NRI Junior College – Intermediate (M.P.C) (Percentage: 75%)

Guntur, India – March 2022

Z.P. High School – Secondary Education (SSC) (Percentage: 93%)

Guntur, India – March 2020

Certifications

Introduction to Large Language Models – Simplilearn

July 2026

Full Stack Web Development Course – Aajhub & Sapienz Recruit

May 2025

Complete AWS Course – Aajhub & Sapienz Recruit

May 2025

Artificial Intelligence – Infosys Springboard

January 2025

Java Programming – Infosys Springboard

June 2024

Introduction to SQL – Simplilearn

July 2023

Python Bootcamp – Dev Nirwal

April 2022